



FACULTY OF COMPUTING

SEMESTER 2 2025/2026

SECP3133 - HIGH PERFORMANCE DATA PROCESSING

SECTION 2

LECTURER: DR. SEAH CHOON SEN

WEEK 4 - EXERCISE

TITLE:

CASE STUDY - TESLA

GROUP

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Part A: Case Study Analysis

A1. Data Characteristics

The Tesla autonomous driving ecosystem relies on a highly complex, hybrid data architecture to manage the flow of vehicle telemetry and environmental data. The system involves two primary data types. The first is real time streaming data, which is responsible for immediate vehicle navigation, obstacle detection, and collision avoidance. Batch data is used for non-critical operations, such as uploading daily vehicle health logs, aggregated battery metrics, and driver intervention logs. It is normally uploaded in batches when the vehicle connects to the internet.

The data used in Tesla's autonomous driving system comes from a combination of onboard sensors, vehicle systems, and user inputs. Primary sources include multiple cameras that provide visual coverage up to 250 meters, a forward-facing radar that helps detect objects and measure distance in various weather conditions, and 12 long-range ultrasonic sensors placed around the vehicle to sense nearby objects within approximately 16 feet. These sensors are providing real-time environmental and spatial data. The system also utilises data from the vehicle's digital electric assist braking system, driver input and system logs. Other inputs may include fleet-level data, user patterns and system events, which are used to train machine learning algorithms that enhance navigation, object recognition, and other aspects of autonomous driving.

Three dimensions of data:

- **Volume:** A global millions of Tesla vehicles produces massive amounts of data, ingesting and storing petabytes daily from cameras, radar, ultrasonic sensors, and vehicle systems. This scale requires moving beyond traditional relational databases into massive, distributed cloud storage environments.
- **Velocity:** The data is produced and processed at extremely high speeds. The vehicle's onboard computer must handle real-time streaming data to enable immediate decision-making for safe navigation. At the same time, Tesla's cloud infrastructure continuously ingests high speed telemetry streams from millions of concurrent vehicle connections.
- **Variety:** The system processes a wide range of data formats. It handles massive amounts of unstructured data such as High Definition video and image frames alongside structured and semi-structured data such as JSON-formatted telemetry, timestamped GPS coordinates, and time-series sensor logs.

A2. Key Challenges

Latency: The biggest challenge is the ability to process information from the environment at near zero latency. Autonomous vehicles can not wait for the cloud to make a decision. If there is network latency between transmitting a video stream from a car to the cloud, it could lead to an accident as receiving a command to brake is too long.

Data Processing Speed & Scalability: The system is highly scalable. So, the cloud data centre must be able to ingest and manage multiple, high velocity data streams from millions of vehicles over different network conditions either 5G or LTE without loss of packets and without facing problem in the ingestion layer.

Data Retention and Cost: Data retention and costs to store data are a challenge. It is technically and economically impossible to store petabytes of raw 4K video data from all vehicles. The system can intelligently process and compress data at the edge and upload only the useful data to the central data store.

A3. Technologies Required

AI and Machine Learning Tools: The system uses a deep learning ecosystem to handle real time prediction and classification.

- **Edge Inference:** Tesla utilizes its custom-built Full Self-Driving Chip optimized for maximum performance-per-watt to run AI inference locally. Instead of relying on the internet to make decisions, the car does all its thinking locally. By processing everything inside the car, the system can react to hazards in milliseconds, rather than waiting for a signal to go back to a server.
- **Neural Networks:** The system deploys 48 different deep neural networks that output 1000 predictions per timestep. Per-camera networks perform complex classification tasks like semantic segmentation and object detection, while "birds-eye-view" networks combine video from all cameras to output a 3D map of the car's surroundings.
- **Autonomy Algorithms:** The neural networks gather all the information and take all the visual data and turn it into actual driving actions. They calculate the safest path forward and help the car make smart decisions even in messy or unpredictable driving conditions.

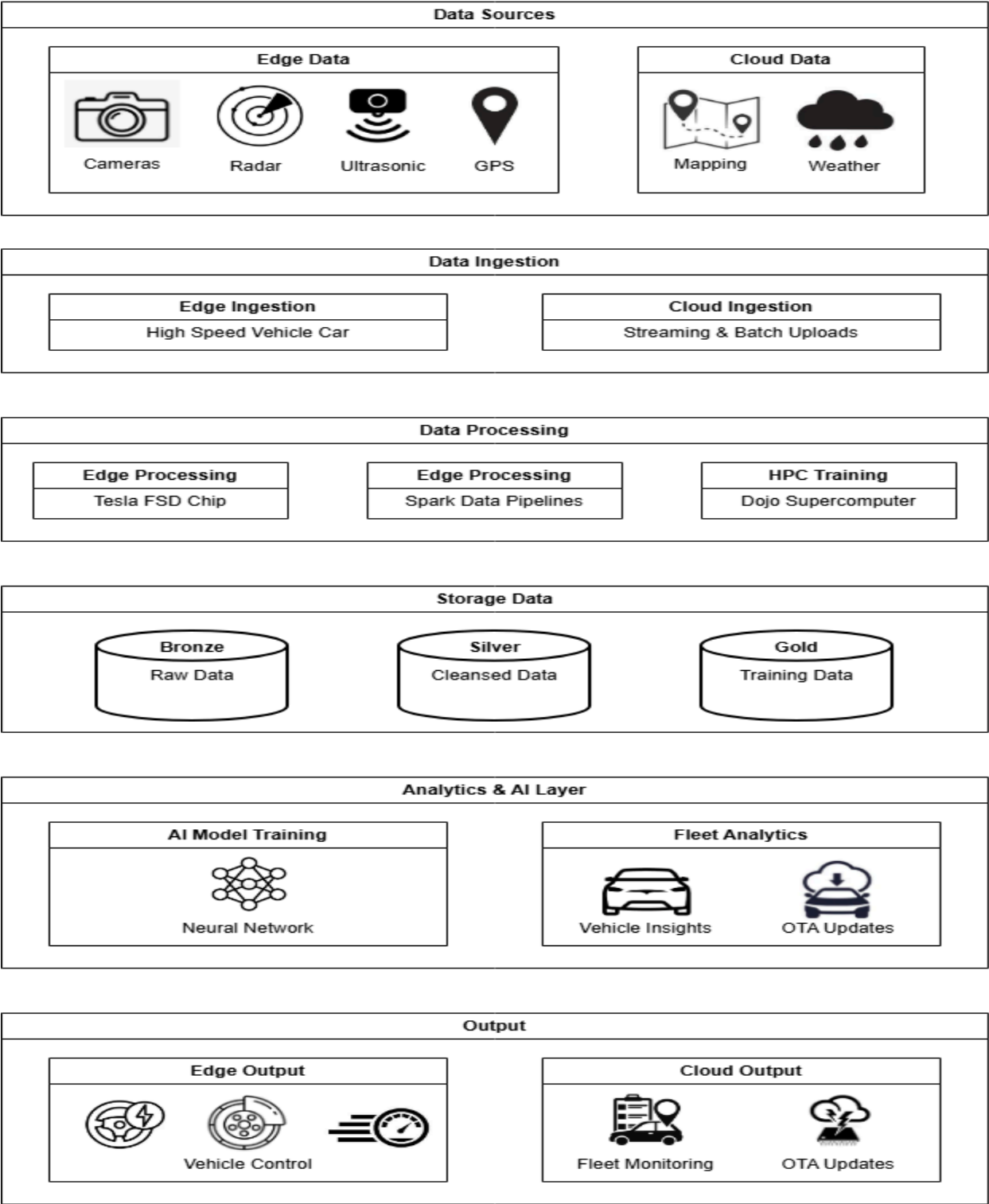
Streaming Platforms: Tesla uses tools like Apache Kafka that acts as a traffic controller to handle fast and continuous flow of data coming from millions of cars at the same time. It helps

collect the massive amount of live vehicle data safely and smoothly deliver it to the next stage for processing without overloading the system.

Storage Systems: A Data Lakehouse is used to store large and different types of data. The system uses Databricks to organise the data into three simple steps. First, raw data is saved exactly as it comes in from the car (Bronze layer). Next, the data is filtered and cleaned up (Silver layer). Finally, it is neatly organized so the AI can quickly use it for training and analysis (Gold layer).

Part B: Architecture Design

Tesla Autonomous Driving System Architecture



Part C: Justification

C1. Architecture Justification

The proposed architecture can be used for Tesla because it can support real-time decision-making in the vehicle, as well as large-scale processing of fleet data. The system receives information from data sources such as cameras, GPS, vehicle logs, maps, and weather data, which help it understand road conditions and how the vehicles are being driven.

Edge ingestion and edge processing enable data to be processed inside the vehicle with minimal latency. This is crucial because safety-related operations such as obstacle detection, lane detection, braking, and steering need to occur instantly. Selected vehicle data is ingested and processed in the cloud so that it can be cleaned and analysed at scale.

The lakehouse storage design is divided into three layers: Bronze, Silver, and Gold. The data is initially stored, then cleaned, and then prepared for model training and analytics. The Analytics and AI layer enables neural network training, fleet analytics, and system improvement. The final product will feature vehicle control, fleet monitoring, and over-the-air updates, enabling Tesla to continually refine its driving system.

C2. Computing Paradigm Justification

This architecture relies on a combination of edge computing, cloud computing, and HPC. Edge computing is employed due to the need to make driving decisions locally within the vehicle without waiting for the cloud. This helps to minimize delay and enhance safety.

Big data gathered from numerous vehicles is stored, processed, and analysed in the cloud. It also provides fleet monitoring and over-the-air software updates. For training deep learning models, HPC is used due to the requirements of autonomous driving for heavy computation on large video and sensor datasets.

Overall, edge computing provides fast real-time response, cloud computing provides scalability, and HPC provides high processing power for AI training. They combine to make the architecture appropriate for Tesla's autonomous driving system.

Part D: Presentation

References

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